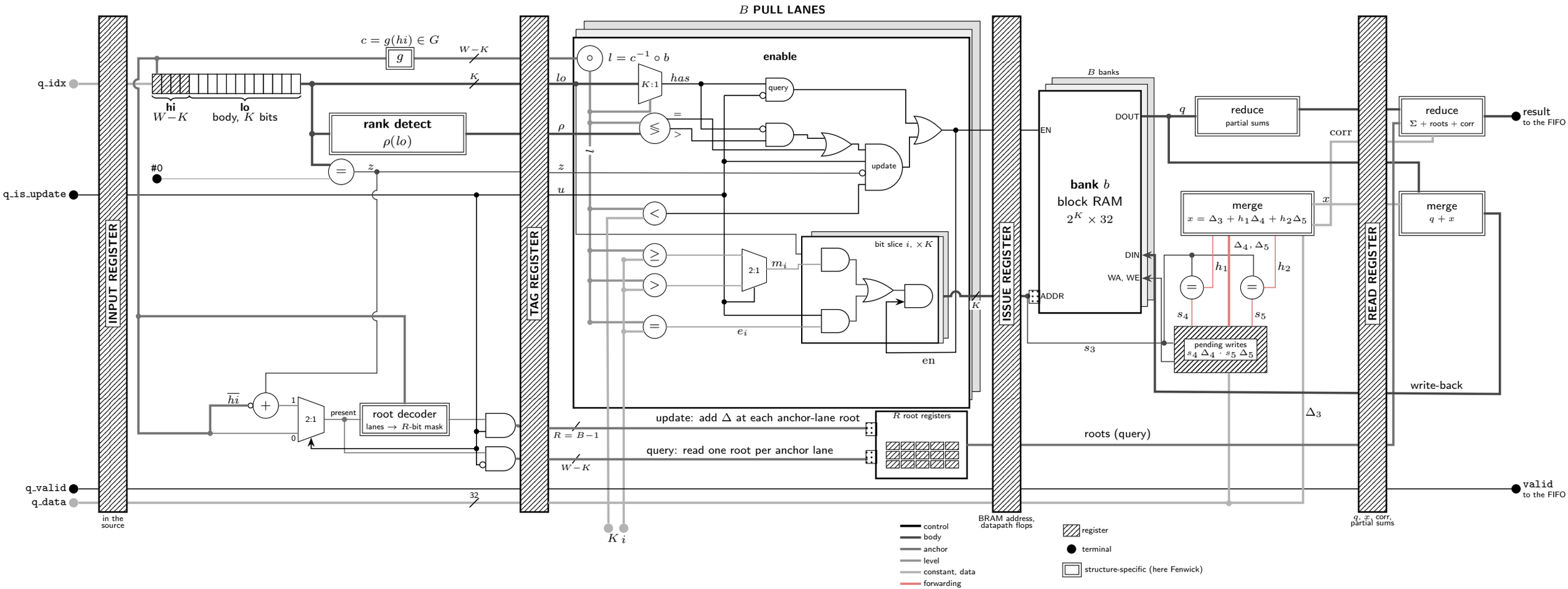

B PULL LANES
q_idx
q_is_update
q_valid
q_data
in the source
INPUT REGISTER
$c = g(hi) \in G$
g
$W-K$
K
hi
$W-K$
lo
body, K bits
rank detect
$\rho(lo)$
#0
z
$\overline{hi}$
1
0
2:1
present
root decoder
lanes → R-bit mask
32
TAG REGISTER
lo
ρ
z
u
$l = c^{-1} \circ b$
l
K:1
has
enable
query
update
m_i
bit slice i, $\times K$
e_i
en
K
K i
$R = B-1$
$W-K$
update: add Δ at each anchor-lane root
query: read one root per anchor lane
R root registers
ISSUE REGISTER
BRAM address, datapath flops
B banks
DOUT
EN
bank b
block RAM
$2^K \times 32$
DIN
WA, WE
ADDR
s_3
roots (query)
q
reduce
partial sums
corr
merge
$x = \Delta_3 + h_1\Delta_4 + h_2\Delta_5$
x
Δ_4, Δ_5
h_1
h_2
s_4
s_5
pending writes
$s_4\,\Delta_4 \cdot s_5\,\Delta_5$
Δ_3
READ REGISTER
q, x, corr, partial sums
reduce
Σ + roots + corr
result
to the FIFO
merge
$q + x$
write-back
valid
to the FIFO
control
body
anchor
level
constant, data
forwarding
register
terminal
structure-specific (here Fenwick)